Jacob Harris

00JacobHarris@gmail.com US Citizen

PROFILE

Aerospace engineering student graduating May 2027 with aircraft loads, stress analysis, computer aided design, and finite element modeling experience. Seeking an internship or part-time role in aerospace structures.

EDUCATION

California State Polytechnic University, Pomona CA (CPP) **Exp. Grad May 2027**

B.S. Aerospace Engineering | ***Materials Science Engineering Minor***

- ***Related Coursework:*** *Systems Engineering, Aerospace Structures 1 & 2, Aerospace Structures Lab, Material Science Lab, Failure and Fracture Analysis, Finite Element Analysis (FEA), Vibrations, Ethical Concepts in Applied Science, Wind Tunnel Lab(High and Low speed wind tunnels)*

SKILLS

MSC NASTRAN, Patran, FEMAP, SOLIDWORKS, Siemens AMESim, MATLAB/Simulink, Java, Python, Arduino IDE, Excel, & PowerPoint

PROJECTS

Battery as iNtegrated Structure High Endurance Experimental (BANSHEE) UAV **Aug. 2026 – Present**

Assistant Design & Structures Lead

- Resized landing gear diameter, resolving negative safety margins and achieving a 1.5 FoS under 2G landing loads
- Evaluated 1G and 2G flight and landing load cases; used FEMAP to identify critical load locations in wing/horizontal tail
- Led a SOLIDWORKS workshop, teaching CAD fundamentals and introducing structural mechanics and stress analysis

Maelstrom Tiltrotor Drone eVTOL Project **Aug. 2024 – Aug. 2026**

Design & Structures Lead

- Hand calculated fuselage and wing stresses match with MSC NASTRAN/FEMAP predictions
- Optimized fuselage shape and materials, reducing weight by 15 lb while maintaining a FoS of at least 1.5
- Flight tests verified no structural failures at critical points

AIAA Design Competition – Hybrid-Electric STOL **Aug. 2026 – Present**

Team Lead

- Iteratively updated aircraft geometry, component weight, CG, L/D, and battery sizing to abide by RFP requirements
- Developed SOLIDWORKS aircraft geometry from calculations for wings, fuselage, nacelles, and landing gears
- Evaluated weight distribution and CG location for stability and control; conducted engine and motor trade studies to guide propulsion selection

F-14 Tomcat Conceptual & Preliminary Design Study – Systems Engineering **Aug. 2024 – Dec. 2024**

Team Lead

- Coordinated project schedules, task tracking, and review preparation for an F-14 conceptual and preliminary design study
- Tested F-14 model in CPP's low-speed wind tunnel, generated L/D curves, and presented finding at PDR

PUBLICATIONS

Harris, J., Saunders, G., Sumagaysay, G., Patel, S., Iribe, A., Francis, D., Salib, G., Escalante, J., Amaya, K., Sherman, T. M., and Bhandari, S., "Design, Analysis, and Testing of a Modular Tiltrotor eVTOL Vehicle with Distributed Electric Propulsion," *AIAA Paper* **2026-1878**, 2026. https://doi.org/10.2514/6.2026-1878

ADDITIONAL WORK EXPERIENCE – Server Assistant, Temecula Creek Inn Oct. 2021 – July 2025

MEMBERSHIPS & AFFILIATIONS

American Institute of Aeronautics and Astronautics (AIAA)
Aug. 2025 – Present

LINKEDIN & ENGINEERING PORTFOLIO